

Deep Learning for Predictive Medicine

A Critical Review

A critical analysis of “Advances in AI and Machine Learning for Predictive Medicine”
Sharma, Lysenko, Jia, Boroevich & Tsunoda (2024), *Journal of Human Genetics*

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1. Research Background

Genome-wide association studies (GWAS) have turned up thousands of variants linked to disease risk, but most of them, taken individually, barely move the needle — their effects are small and scattered across the genome. At the same time, multi-omics data (genomics, transcriptomics, proteomics, and so on) keeps piling up faster than most labs can make sense of it with conventional tools. Random forests and SVMs can pick up some of this signal, but they tend to hit a wall when the relationships between features are nonlinear or spread thin across thousands of variables at once.

That's really the gap this review sits in: can deep learning, and CNNs specifically, do a better job of finding structure in this kind of data than the classical toolkit has managed so far?

2. Paper Overview

Paper reviewed: “Advances in AI and Machine Learning for Predictive Medicine” (2024), Alok Sharma, Artem Lysenko, Shangru Jia, Keith A. Boroevich & Tatsuhiko Tsunoda, *Journal of Human Genetics*.

Type: Review article **Area:** Bioinformatics, Machine Learning, Genomics

The review's central argument is that a family of techniques built around **DeepInsight** — a method that converts tabular omics data into image-like representations — lets CNNs (originally built for photographs, not gene expression tables) be applied to genomic and multi-omic prediction problems. Alongside DeepInsight itself, the paper surveys three related tools: DeepFeature (for interpretability), DeepInsight-3D (for multi-omics integration), and scDeepInsight (for single-cell data), and closes by laying out the challenges the approach still hasn't solved.

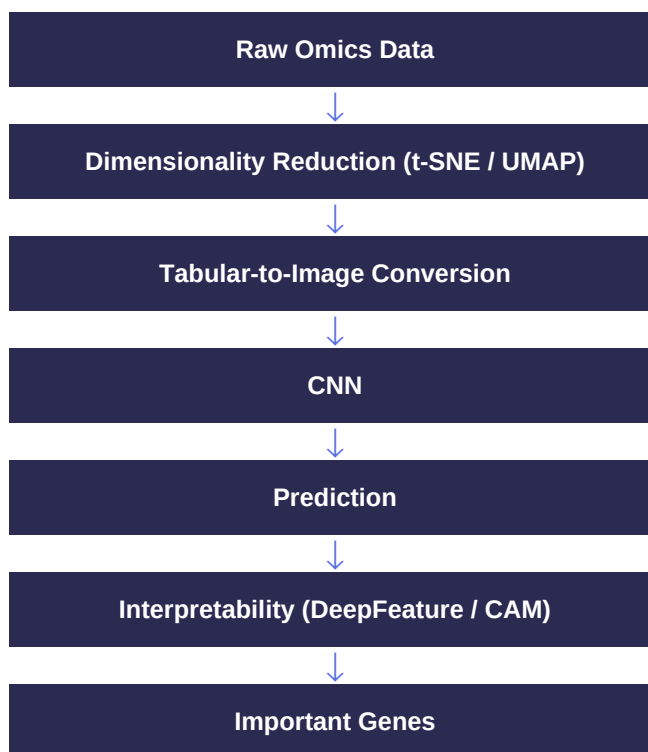
3. Methodological Framework

Since this is a review rather than a paper reporting new experiments, its “method” ends up being more of an explanatory pipeline than a single algorithm. Broken down, it runs roughly like this:

- **Data transformation:** manifold learning (t-SNE, UMAP, or kernel PCA) places each gene as a point in 2D space based on how similar its expression pattern is to other genes, so genes that behave similarly end up sitting close together.
- **Image construction:** a convex hull algorithm finds the smallest bounding rectangle around those points, rotates it, and maps it onto a pixel grid — turning a gene expression vector into an actual image a CNN can read.
- **CNN classification:** the generated images go into a CNN (often pretrained via transfer learning) for tasks like cancer-type classification or drug-response prediction.
- **Interpretability layer:** DeepFeature uses class activation maps (CAMs) to flag which regions of the generated image — and therefore which genes — drove the model's decision.
- **Extensions:** DeepInsight-3D stacks multiple omics layers (mutation, expression, copy number) into a 3D version of the same idea, while scDeepInsight applies it to single-cell RNA-seq data for cell-type annotation.

3.1 Pipeline Overview

Visually, the whole idea compresses into a fairly simple chain:



4. Key Findings

- DeepInsight-3D beat several competing models (feed-forward networks, autoencoders, random forest pipelines, SVM-based classifiers, and other published drug-response pipelines) by 7–29% in AUC-ROC for predicting anti-cancer drug response.
- In a narrower comparison against other deep-learning methods specifically, DeepInsight-3D hit 72% accuracy in drug-response prediction — over 7% better than the next-best deep-learning approach.
- DeepFeature picked out known cancer-related pathways (STAT3, PI3K/AKT, JAK/STAT) and also surfaced one that wasn't previously flagged in this context: tryptophan metabolism.
- scDeepInsight improved single-cell type annotation accuracy by more than 7% over existing methods, and the authors suggest it may help identify rare or previously unrecognized cell types.
- Transfer learning from unrelated image datasets like ImageNet noticeably boosted performance on small omics datasets — directly addressing the data-scarcity problem that plagues a lot of genomics work.

4.1 Results Summary

Method	Task	Performance	Compared Against
DeepInsight-3D	Drug response prediction	72% accuracy	Other deep-learning methods (+7%)
DeepInsight-3D	Drug response prediction	7–29% higher AUC-ROC	Random forests, SVMs, autoencoders, published pipelines
DeepFeature	Gene interpretation	Identified STAT3, PI3K/AKT, JAK/STAT	Improves model interpretability
scDeepInsight	Cell type annotation	>7% improvement	Better identification of rare cell types
Transfer Learning	Small omics datasets	Improved prediction performance	Helps overcome limited training data

Note: the two DeepInsight-3D rows come from two different comparison sets in the original paper — one benchmark against deep-learning methods specifically (72% accuracy), and a broader one against classical ML and published pipelines (7–29% AUC-ROC gain). They aren't duplicate measurements of the same test.

4.2 Model Contributions

Model	Main Result	Contribution to Bioinformatics
DeepInsight	Converts tabular omics data into images	Enables CNNs to analyze genomic data
DeepFeature	Highlights influential genes	Improves model interpretability
DeepInsight-3D	Integrates multi-omics data	Supports more accurate drug-response prediction
scDeepInsight	Analyzes single-cell RNA-seq	Improves cell-type annotation

(Performance metrics summarized from the reviewed paper; they are not results generated by this analysis.)

What sticks with me most from this section isn't any single number, it's that the same trick — wrap tabular data in image form — keeps paying off across genuinely different problems: bulk drug response, single-cell typing, pathway discovery. That kind of reuse is usually a decent signal that an idea is doing real work, not just fitting one dataset well.

5. Critical Analysis

5.1 Strengths

- Innovative tabular-to-image representation
- Effective use of transfer learning
- Improved interpretability through CAMs

5.2 Weaknesses

- Limited validation on diverse datasets
- Computationally expensive
- Biological information may be lost during image conversion

5.3 Advantages vs. Limitations, at a Glance

Advantage	Corresponding Limitation
Reuses mature CNN architectures and pretrained weights	Computationally expensive to train and tune at scale
Effective on small omics datasets via transfer learning	Still needs a reasonably sized dataset for a stable image mapping
CAMs make predictions traceable back to specific genes	Interpretability is tied to the image layout, not the biology directly
Works across bulk, multi-omics, and single-cell data	Spatial placement (t-SNE/UMAP) may not reflect true biological relationships

5.4 CNN-Based Image Conversion vs. Graph Neural Networks

One comparison the review doesn't really make, but that I kept coming back to, is how this approach stacks up against graph neural networks (GNNs) — a different way of handling relationships between genes that's gained a lot of traction in bioinformatics recently.

Aspect	CNN + Image Conversion (DeepInsight family)	Graph Neural Networks
How relationships are captured	Inferred indirectly from t-SNE/UMAP spatial proximity	Encoded explicitly as edges from a known gene-interaction network
Reuse of existing tooling	Can reuse standard image-based CNN architectures and pretrained weights	Needs graph-specific architectures (e.g., GCN, GAT)
Biological grounding	Not guaranteed — proximity in the image may not reflect real biology	Can be grounded directly in curated biological interaction data
Data requirements	Needs enough samples for a stable 2D projection	Needs a reliable interaction network, which may itself be incomplete
Interpretability	CAMs highlight image regions / genes	Attention weights or edge importance can highlight specific interactions

5.5 My Perspective

The part of this pipeline I'm most skeptical of is the image conversion step itself. Positioning genes by t-SNE or UMAP proximity captures similarity in expression pattern across samples, but it doesn't preserve anything like a known regulatory network or pathway hierarchy. Two genes can end up next to each other in the generated image simply because of how the projection happened to fold the data that particular run, not because they interact biologically in any established sense. That matters for more than just correctness: if the interpretability story (DeepFeature, CAMs) is meant to build clinical trust by tying predictions back to “this gene mattered,” it's worth knowing whether the gene's position in the image reflects a real biological relationship or an artifact of the dimensionality-reduction step.

Graph neural networks sidestep this specific issue by encoding actual gene-gene interaction networks as edges, rather than inferring spatial closeness after the fact. That feels like a more principled way to keep biological structure intact, even though GNNs bring their own dependency — they're only as good as the interaction network they're built on, which is itself incomplete for many diseases. I'd want to see DeepInsight-3D and scDeepInsight benchmarked directly against a GNN baseline (something like Liu et al.'s MTIGCN, a graph convolutional network built specifically for drug-response prediction) on the same datasets, rather than concluding image-based CNNs are the better long-term route just because they let researchers reuse existing CNN tooling. Convenience and correctness aren't the same thing, and right now it's hard to tell which one is driving DeepInsight's adoption.

6. Limitations

The authors are fairly upfront that several problems remain open: CNN-based omics models still need large datasets despite what transfer learning buys them, results can get unstable when training data is limited (explicitly flagged for DeepInsight-3D), and there's no guarantee that biological meaning survives the tabular-to-image conversion intact. The review also concedes that benchmarking across different methods and datasets is inconsistent right now, which makes it genuinely hard to compare approaches on equal footing.

6.1 Traditional ML vs. DeepInsight

Traditional ML	DeepInsight
Manual feature engineering	Automatic image representation
Random Forest / SVM	Convolutional Neural Network
Features treated independently	Spatial relationships between features

7. Future Directions

The authors expect tabular-to-image methods to keep integrating more omics layers at once, fold in more domain-specific biological knowledge during training, and move toward real-time clinical applications where genomic data could inform treatment decisions on the fly. They also flag continued work on interpretability tools as essential if any of this is going to earn clinical trust.

8. Lessons Learned

Going into this paper, I hadn't really thought about computer vision and genomics as fields that could borrow from each other so directly. Seeing gene expression data get reshaped into literal images, just so a CNN built for photographs could be pointed at it, was the part that reframed how I think about model architecture choices in general: sometimes the more valuable innovation isn't a new network design, it's a new way of representing the data so an existing, well-understood architecture becomes usable.

It also sharpened how I think about explainability. Before this, I'd mostly encountered "interpretability" as a box to check rather than something load-bearing. Here it's doing real work — DeepFeature's CAMs are the mechanism that turns a black-box prediction into something a biologist could actually act on, which is a big part of why this kind of method could plausibly reach clinical use rather than staying a benchmark exercise.

Finally, this pushed small-data learning higher up my list of problems worth caring about. A lot of the more exciting deep learning results I'd read about before assumed large, clean datasets, and rare-disease genomics almost never has that luxury. Transfer learning from something as unrelated as ImageNet actually helping here was a genuinely useful thing to learn, not just a footnote.

9. Potential Research Extensions

A few directions feel worth pursuing after sitting with this paper for a while:

- Compare DeepInsight against transformer-based models for multi-omics analysis.
- Evaluate graph neural networks for preserving biological relationships that image conversion may lose.
- Test DeepInsight on publicly available RNA-seq datasets from TCGA and GEO.
- Explore lightweight DeepInsight variants for hospitals or labs with limited computational resources.

10. Questions I Still Have

Reading this raised as many questions as it answered, which I think is a reasonable sign that I engaged with it rather than just skimming for the results section:

- Would a graph neural network actually outperform image-based CNNs on the same drug-response task, or is the CNN's advantage mostly about being able to reuse pretrained weights?
- Does the t-SNE/UMAP step introduce instability between runs — would the same dataset produce a meaningfully different image layout, and different CAM explanations, on a second pass?

- Would transformer-based architectures remove the need for image conversion entirely, or would they just replace one representation problem with another?
- How well would any of this hold up on rare disease datasets, where sample sizes are far smaller than even the cancer datasets used here?
- If the image layout changes between training runs, how much does that undermine DeepFeature's gene-level explanations for a clinician trying to act on them?

11. How This Paper Influences My Future Projects

This review has pushed a few concrete project ideas higher on my list. I'd like to try applying the DeepInsight approach to a public RNA-seq dataset from TCGA or GEO myself, mostly to get a feel for where the image conversion step actually breaks down in practice rather than just reading about it. I'm also interested in setting up a small side-by-side comparison between an image-based CNN and a graph-based model on the same drug-response prediction task, since that's the open question this review left me with rather than one it answered.

More broadly, this shifted how I want to approach explainable AI in any future multi-omics project: rather than treating interpretability as an add-on at the end, I want to think about it as part of the core design from the start, the way DeepFeature is built into this pipeline rather than bolted onto it afterward.

12. Conclusion

DeepInsight's core idea — that a CNN doesn't actually care whether the pixels in front of it came from a camera or from a gene expression table — is the most useful takeaway from this review. It reads less like a fundamentally new theory of biology and more like a practical workaround for a real bottleneck: small, high-dimensional omics datasets that don't play well with standard deep learning. I think that's a fair thing to be, and the drug-response and cell-typing results suggest it's earning its keep on real tasks rather than surviving purely as a clever trick. Whether it stays the default approach as GNN and transformer-based methods mature is still an open question, and one the field doesn't yet have a consistent enough benchmarking standard to answer cleanly.

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